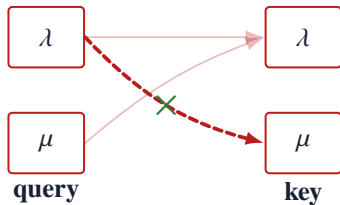
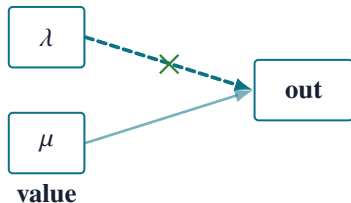


Feature-Resolved Attention

QK: which feature pair



OV: which feature moves



$$\text{FRA}_{qk,\lambda\mu}^{\text{QK},h} = \frac{1}{\sqrt{d_h}} f_q^\lambda W_{QK}^h (f_k^\mu)^\top$$
$$f_q^\lambda \equiv u_q^\lambda W_\lambda^{\text{dec}}$$

$$\text{FRA}_{k,\lambda}^{\text{OV},h} = W_{OV}^h f_k^\lambda$$